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MANIPAL SCHOOL OF INFORMATION SCIENCES
(A Constituent unit of MAHE, Manipal)

Linux OS and Scripting – Python

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How is Python programming useful in cybersecurity? Explain any two applications of Python in cybersecurity, with a suitable example for each.

Python is useful in cybersecurity because it can automate repetitive tasks, process large amounts of data, and work with files and network tools. This helps security professionals detect suspicious activity and protect systems more efficiently.

Network scanning: Security professionals can use Python tools to identify active devices, open ports, and running services on networks they are authorized to assess. This information can help administrators find and fix security weaknesses.

Malware analysis: Security researchers can use Python to examine suspicious files and study their behaviour in a controlled environment. This can help them understand how malware works and develop ways to detect it.

Analysing security logs: Python can search log files for patterns that may indicate attacks, such as repeated failed logins.

Example: A script could count failed-login messages by IP address. If one address appears many times, an administrator can investigate whether it is attempting to guess passwords.

```
from collections import Counter

# Sample log entries
logs = [
    "Failed login from 192.168.1.10",
    "Successful login from 192.168.1.20",
    "Failed login from 192.168.1.10",
    "Failed login from 192.168.1.15",
    "Failed login from 192.168.1.10"
]

failed_attempts = Counter()

for line in logs:
    if "Failed login from" in line:
        ip_address = line.split()[-1]
        failed_attempts[ip_address] += 1

for ip_address, count in failed_attempts.items():
    print(ip_address, "had", count, "failed login attempts")
    if count >= 3:
        print("  Check this address: several failed attempts detected.")
```

Checking file integrity: Python can calculate a file's hash, a value that changes if the file's contents change. Comparing hashes over time can reveal unexpected modifications.

Example: An administrator records the SHA-256 hash of an important system file. Later, a Python script calculates it again and alerts them if the value differs, which may indicate unauthorized changes.
These applications help teams monitor systems and respond to potential security problems

```python
import hashlib

file_name = "important_file.txt"

try:
    with open(file_name, "rb") as file:
        contents = file.read()

    file_hash = hashlib.sha256(contents).hexdigest()
    print("SHA-256 hash:", file_hash)

except FileNotFoundError:
    print("The file was not found.")
```

Overall, Python helps make cybersecurity work more efficient and supports the detection and investigation of potential security problems.